LP-3 Practical Viva Notes

DAA | ML | BCT - short and simple answers

Subject 1: DAA (Design and Analysis of Algorithms)

Practical 1: Fibonacci Series (Recursive and Non-Recursive)

Q1. What is the Fibonacci series?

Ans. It is a sequence in which each number is the sum of the previous two. Example: 0, 1, 1, 2, 3, 5, 8.

Q2. What is recursion?

Ans. Recursion is when a function calls itself.

Q3. What is non-recursion?

Ans. Solving a problem without calling the same function again.

Q4. What is the base condition?

Ans. The condition that stops the recursive function.

Q5. Which method is faster for Fibonacci?

Ans. The iterative (non-recursive) method is generally faster.

Q6. Time complexity of recursive Fibonacci?

Ans. About $O(2^n)$ for the simple recursive method.

Q7. Time complexity of iterative Fibonacci?

Ans. O(n).

Practical 2: Huffman Encoding

Q1. What is Huffman encoding?

Ans. A compression technique used to reduce the size of data.

Q2. Which algorithm is used?

Ans. Greedy algorithm.

Q3. What is a greedy algorithm?

Ans. It picks the best available option at each step.

Q4. What is a Huffman tree?

Ans. A binary tree used to generate Huffman codes.

Q5. Which character gets the shortest code?

Ans. The character with the highest frequency.

Q6. What is Huffman encoding used for?

Ans. Data compression.

Practical 3: Fractional Knapsack

Q1. What is the Knapsack problem?

Ans. Selecting items to get maximum profit within a given capacity.

Q2. What is fractional knapsack?

Ans. A version where we can take a part of an item.

Q3. Which algorithm is used?

Ans. Greedy algorithm.

Q4. How do we select items?

Ans. By their profit-to-weight ratio, highest first.

Q5. Formula for profit-to-weight ratio?

Ans. Profit divided by weight.

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Q6. Can we divide items in fractional knapsack?
Ans. Yes.

Practical 4: 0/1 Knapsack

Q1. What is 0/1 Knapsack?
Ans. A problem where we either take the complete item or leave it.

Q2. Can we divide items in 0/1 knapsack?
Ans. No.

Q3. Which technique is used?
Ans. Dynamic Programming.

Q4. What is dynamic programming?
Ans. Solving a problem by breaking it into smaller subproblems and storing their results.

Q5. What do 0 and 1 mean?
Ans. 0 means the item is not selected, 1 means it is selected.

Q6. What is the time complexity?
Ans. O(nW), where n is the number of items and W is the capacity.

Practical 5: N-Queens Problem

Q1. What is the N-Queens problem?
Ans. Placing N queens on an N x N chessboard so that no queen attacks another.

Q2. Which technique is used?
Ans. Backtracking.

Q3. What is backtracking?
Ans. Trying a solution and going back if it is not correct.

Q4. How does a queen move?
Ans. Horizontally, vertically and diagonally.

Q5. Can two queens be in the same row?
Ans. No.

Q6. Can two queens be in the same column?
Ans. No.

Q7. What is the use of the N-Queens problem?
Ans. It helps us understand backtracking and constraint-solving problems.

Subject 2: ML (Machine Learning)

Practical 1: Uber Fare Prediction

Q1. What is Machine Learning?
Ans. A part of AI that lets computers learn from data.

Q2. What is fare prediction?
Ans. Predicting the taxi fare using available data.

Q3. Which algorithms are used?
Ans. Linear Regression and Random Forest Regression.

Q4. What is Linear Regression?
Ans. An algorithm used to predict continuous values.

Q5. What is Random Forest?
Ans. It uses multiple decision trees to make predictions.

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Q6. What is preprocessing?
Ans. Cleaning and preparing data before training.

Q7. What is an outlier?
Ans. A value that is very different from the other values.

Q8. What is the R^2 score?
Ans. It tells us how well the model explains the data.

Q9. What is RMSE?
Ans. It measures the prediction error of a model.

Practical 2: Email Spam Detection

Q1. What is spam email?
Ans. An unwanted or unnecessary email.

Q2. What is email spam detection?
Ans. Identifying whether an email is spam or not spam.

Q3. Which algorithms are used?
Ans. KNN and SVM.

Q4. What is KNN?
Ans. K-Nearest Neighbors. It predicts results using nearby data points.

Q5. What is SVM?
Ans. Support Vector Machine. It separates data into different classes.

Q6. What is classification?
Ans. Dividing data into categories.

Q7. What is binary classification?
Ans. Classifying data into two categories, such as spam and not spam.

Practical 3: Bank Customer Churn Prediction

Q1. What is customer churn?
Ans. When a customer stops using a company's services.

Q2. What is churn prediction?
Ans. Predicting whether a customer will leave the bank or stay.

Q3. Which model is used?
Ans. Neural Network.

Q4. What is a Neural Network?
Ans. A machine learning model inspired by the human brain.

Q5. What is a neuron?
Ans. The basic processing unit of a neural network.

Q6. What is training data?
Ans. The data used to teach the model.

Q7. What is prediction?
Ans. Estimating the result using a trained model.

Practical 4: Diabetes Prediction

Q1. What is diabetes prediction?
Ans. Predicting whether a person may have diabetes from the given dataset.

Q2. Which algorithm is used?
Ans. K-Nearest Neighbors (KNN).

Q3. What is a confusion matrix?
Ans. A table used to check the performance of a classification model.

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Q4. What is accuracy?
Ans. How many predictions are correct.

Q5. What is precision?
Ans. Of all predicted positives, how many are actually positive.

Q6. What is recall?
Ans. Of all actual positives, how many were correctly identified.

Q7. What is error rate?
Ans. How many predictions are incorrect.

Q8. What is KNN?
Ans. It predicts the class of new data using its nearest data points.

Practical 5: Sales Clustering

Q1. What is clustering?
Ans. Grouping similar data points together.

Q2. Which algorithms are used?
Ans. K-Means and Hierarchical Clustering.

Q3. What is K-Means?
Ans. An algorithm that divides data into K groups.

Q4. What is a centroid?
Ans. The center point of a cluster.

Q5. What is the Elbow Method?
Ans. It helps us choose a suitable number of clusters.

Q6. What is unsupervised learning?
Ans. Machine learning on data that has no labelled outputs.

Q7. What is the use of sales clustering?
Ans. It groups customers by similar purchasing behaviour.

Subject 3: BCT (Blockchain Technology)

Practical 1: MetaMask Installation and Ether Spending

Q1. What is Blockchain?
Ans. A digital record system that stores data in connected blocks.

Q2. What is MetaMask?
Ans. A cryptocurrency wallet used to interact with blockchain networks.

Q3. What is cryptocurrency?
Ans. A digital currency that uses cryptography.

Q4. What is Ether?
Ans. Ether (ETH) is the cryptocurrency used on the Ethereum network.

Q5. What is a gas fee?
Ans. The fee paid to perform transactions on the Ethereum network.

Q6. What is a transaction?
Ans. An operation performed on a blockchain network.

Practical 2: Create a Wallet Using MetaMask

Q1. What is a crypto wallet?
Ans. A tool used to manage cryptocurrency and blockchain accounts.

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Q2. What is a public key?
Ans. It identifies an account and is used to receive cryptocurrency.

Q3. What is a private key?
Ans. A secret key used to control and access the wallet.

Q4. What is a wallet address?
Ans. A unique address used to send and receive cryptocurrency.

Q5. What is a seed phrase?
Ans. A set of secret words used to recover a wallet.

Q6. Why should we not share the seed phrase?
Ans. Anyone who knows it can get access to the wallet.

Practical 3: Smart Contract for Bank Account

Q1. What is a smart contract?
Ans. A program stored and executed on a blockchain.

Q2. Which language is used?
Ans. Solidity.

Q3. What is Remix IDE?
Ans. An online tool used to write, compile and deploy Solidity smart contracts.

Q4. What is deployment?
Ans. Publishing a smart contract on a blockchain network.

Q5. Functions in our bank smart contract?
Ans. Deposit, Withdraw and Show Balance.

Q6. What is the Deposit function?
Ans. It adds money to the account balance.

Q7. What is the Withdraw function?
Ans. It subtracts money from the account balance.

Q8. What is the Show Balance function?
Ans. It displays the current account balance.

Q9. What is a test network?
Ans. A blockchain network used for testing without real cryptocurrency.

Practical 4: Survey of Blockchain Types and Applications

Q1. What are the types of blockchain?
Ans. Public, Private, Consortium and Hybrid.

Q2. What is a Public Blockchain?
Ans. One that is open to everyone.

Q3. Give examples of Public Blockchain.
Ans. Bitcoin and Ethereum.

Q4. What is a Private Blockchain?
Ans. One controlled by a single organization.

Q5. What is a Consortium Blockchain?
Ans. One managed by a group of organizations.

Q6. What is a Hybrid Blockchain?
Ans. One that combines features of public and private blockchains.

Q7. Real-life applications of blockchain?
Ans. Banking, healthcare, supply chain, digital identity and cryptocurrency.

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Q8. What is decentralization?

Ans. Control is not held by only one central authority.

Practical 5: Hyperledger Business Network

Q1. What is Hyperledger?

Ans. An open-source project used to develop enterprise blockchain solutions.

Q2. What is Hyperledger Fabric?

Ans. A permissioned blockchain platform used by organizations.

Q3. What is Hyperledger Composer?

Ans. A tool used to model and develop business networks on Hyperledger Fabric.

Q4. What is an asset?

Ans. Something valuable that is stored in a business network.

Q5. What is a participant?

Ans. A person or organization involved in a business network.

Q6. What is a transaction?

Ans. An operation that changes the state of an asset or the network.

Q7. What is a .cto file?

Ans. It defines the assets, participants and transactions of a business network.

Q8. What is a BNA file?

Ans. Business Network Archive. It contains the business network files.

Q9. What is an identity?

Ans. It represents a participant in the blockchain network.

Q10. What is the use of Hyperledger?

Ans. It is used to build secure business applications.

Important full forms

Short form	Full form
DAA	Design and Analysis of Algorithms
ML	Machine Learning
BCT	Blockchain Technology
AI	Artificial Intelligence
KNN	K-Nearest Neighbors
SVM	Support Vector Machine
CNN	Convolutional Neural Network
IDE	Integrated Development Environment
ETH	Ether
EVM	Ethereum Virtual Machine
RMSE	Root Mean Square Error
BNA	Business Network Archive

Tip

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For each practical, first prepare the definition, the algorithm or tool used, the purpose and the output. You don't need to memorise the answers word for word. Understand them and say them in your own simple English.

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